Hariprashad Ravikumar

PhD Candidate specializing in High-Performance Computing, Deep Learning, and AI for Science
Expertise in GPU-accelerated computing with C++/CUDA and Python

Website: hariprashad-ravikumar.github.io
Email: hari1729@nmsu.edu
LinkedIn: linkedin.com/in/hariprashad-ravikumar
Phone: +1 575-249-9610
GitHub: github.com/Hariprashad-Ravikumar
Location: San Jose, CA

Experience

1. Incoming **Modeling and Simulation Intern**, Western Digital, San Jose, CA *(May 2026 - Aug 2026)*
2. **Graduate Research Assistant**, NM State University, Las Cruces, NM *(Aug 2021 - Present)*

   PhD Project: Lattice Quantum Chromodynamics & Machine Learning Approaches
   - Generated 30,000+ high-fidelity synthetic data points by solving Partial Differential Equations with large-scale Hybrid Monte Carlo simulations (Markov Chain stochastic process) and built an end-to-end AI for Science pipeline to model the underlying physics, achieving over 98% predictive accuracy with symbolic regression machine learning.
   - Engineered parallelized Bayesian inference pipelines across multi-CPU architectures to fit non-linear, high-dimensional models; utilized automatic differentiation and full covariance matrices to ensure exact error propagation and numerical stability in multi-stage workflows (jackknife resampling)
   - Developed GPU-accelerated CUDA/C++ pipelines for large-scale FFT-based feature extraction, reducing data processing time by 10x

Independent Collaborations

1. **Los Alamos National Laboratory** - Lattice Quantum Chromodynamics *(May 2024 - Present)*
   - Accelerated multi-terabyte scientific calculations by developing and optimizing parallelized C++ CUDA kernels for GPU-accelerated HPC clusters (NERSC Perlmutter), significantly reducing runtime for large-scale Hamiltonian Monte Carlo simulations.
   - Managed and executed 75,000+ CPU/GPU compute hours by designing and deploying custom SLURM workflows for large-scale job orchestration
   - Investigated advanced simulation techniques using gradient flow, a method conceptually similar to Flow-Based Generative Models, to enhance simulation fidelity and ensure numerical stability
   - Increased model reliability through rigorous statistical validation on over 50,000 correlated data points, applying methods like AIC-based selection and chi-squared minimization with full covariance matrices.
2. **North Carolina State University** - Mathematical Physics *(Dec 2020 - Present)*
   - Implemented and managed Mathematica symbolic computation workflows on HPC clusters to analyze complex algebraic structures and symmetry constraints.

Selected Publications

1. C.-R. Ji and **H. Ravikumar**, *"Interpolating conformal algebra in (1+1) dimensions between the instant form and the light-front form of relativistic dynamics,"* **Physical Review D** (March 2026). DOI: 10.1103/prlq-4j1l.

Technical Projects

1. **AI-DataScience-Lab: Full-Stack Data Product** GitHub | Live App
   - Prototyped and deployed a scalable full-stack data tool (Python, Flask, Azure) to deliver on-demand data-driven insights and actionable recommendations, integrating an OpenAI API to demonstrate automated data storytelling and data interpretation.
   - Implemented an end-to-end statistical analysis pipeline using Python (Pandas) for data cleaning, statistical modeling (Scikit-learn) for regression, and Matplotlib for data visualization.
2. **Physics Based Monte Carlo Simulation** GitHub
   - Developed a Physics-Based Simulation from scratch to generate synthetic lattice gauge configurations using Monte Carlo methods on HPC clusters, validating the generated data against known analytical benchmarks.

3. **Neural Network from Scratch with `NumPy`** GitHub
    - Implemented and trained a neural network from scratch in NumPy for MNIST digit recognition, achieving 80% accuracy by building and tuning core components like backpropagation and activation functions

Technical Skills

Programming	Python, C++, CUDA, Bash, SQL, Lua, HTML/CSS, YAML
ML & APIs	Numba, TensorFlow, PyTorch, Scikit-learn, Pandas, cuFFT, cuDNN, Flask, FastAPI, RAG
Cloud & MLOps	Azure, AWS (Lambda, S3), CI/CD, Docker, Git, SLURM, Nsight
Methods & HPC	Parallel Computing (GPU, MPI), Numerical Methods (PDEs, Monte Carlo, Regression)

Education

PhD in Physics, New Mexico State University, USA *Aug 2021 – Dec 2026 (expected)*
MS in Physics, New Mexico State University, USA *Aug 2021 – May 2024*
MSc in Physics, National Institute of Technology Jalandhar, India *July 2019 – May 2021*
BSc in Physics, Dr. N.G.P. Arts and Science College, India *June 2015 – May 2018*

Certifications

- Getting Started with Accelerated Computing in CUDA C/C++ by NVIDIA
- Fundamentals of Accelerated Computing with CUDA Python by NVIDIA
- Advanced Learning Algorithms by DeepLearning.AI
- Supervised Machine Learning: Regression and Classification by DeepLearning.AI
- Google Advanced Data Analytics Professional Certificate

Awards

- **2025 NMC Collaboration Grant**, awarded by the New Mexico Consortium to conduct my independent research project in collaboration with scientists at Los Alamos National Laboratory
- **2023 George and Barbara Goedecke Physics Excellence Fund Scholarship**, awarded by the NMSU Physics Department

Selected Talks

- (Jun 3, 2025) *"First Principles Lattice QCD Calculations of nEDMs"*, T-2 Seminar, Theoretical Division, **Los Alamos National Laboratory**, USA
- (May 16, 2024) *"Lattice QCD Calculations of x Dependence of Sivers TMD"*, T-2 Seminar, Theoretical Division, **Los Alamos National Laboratory**, USA
- (June 15, 2023) *"Lattice QCD calculations of TMDs"*, HUGS Student Seminar, **Thomas Jefferson National Accelerator Facility**, USA

Full list available at: hariprashad-ravikumar.github.io/talks

Volunteering

- **Vice President**, Physics Graduate Student Organization (NMSU) *Aug 2025 – Present*
  Organized professional development events and served as the primary liaison between 40+ graduate students and faculty.

Relevant Graduate Coursework

- Quantum Computing, Advanced Computational Physics, Statistical Mechanics